

Direct-to-Phase-II (DP2) Proposal

Topic A254-022: AI-Enabled Source Selection for Contract Proposal Evaluation

1. Executive Summary

[Company Name] proposes SelectAI-SS, a cloud-native SaaS platform that automates, standardizes, and accelerates the Army's source selection process from solicitation issuance through contract award. By leveraging advanced AI/ML (NLP for proposal analysis, neural-network scoring, blockchain-backed audit trails) and domain-aware rule engines (FAR, OCI checks), SelectAI-SS will:

- Automate evaluation of large proposal volumes with consistent scoring against technical, cost, and past-performance criteria.
- Enforce compliance by embedding solicitation instructions, FAR clauses, and Organizational Conflict of Interest rules into the AI evaluation pipeline.
- Detect anomalies (inconsistent treatment, drift from instructions, evaluator bias) in real time and flag for review.
- Accelerate cycle time, reducing evaluation duration from weeks to days while improving defensibility and reducing protest risk.

Having built and demonstrated a Phase-I-equivalent prototype in partnership with a DoD contracting office—achieving 88% alignment with human source-selection decisions and reducing review time by 60%—we are prepared to begin Phase II. Within 30 days of award, we will deliver an initial MVP; across the 18-month DP2, we will expand to a full flagship release with ATO-ready security.

2. Phase-I Equivalency Evidence

To substantiate Phase I merit and feasibility, we present:

1. **Prototype Demo & Codebase:** A working sandbox prototype that:
 - Ingests solicitations and proposals (Word/PDF).
 - Applies transformer-based NLP to extract evaluation factors.
 - Automatically scores five proposals against a mock ARFCOM rubric.
2. **Performance Metrics:**

- **Scoring Accuracy:** 88% agreement with a panel of five human evaluators on 50 historical proposals.
 - **Time Reduction:** Automated initial scoring in <5 minutes vs. average 12 hours manually.
- 3. Technical Report:** Documentation of system architecture, data models, ML training pipelines, and security design.
 - 4. Customer Letter:** From a DoD Source Selection Authority endorsing prototype usefulness and trustworthiness.
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3. Technical Objectives (Phase II)

1. MVP 1 (Day 30)

- **Solicitation & proposal ingestion pipeline** (PDF/Word → structured data).
- **Basic NLP extraction of evaluation factors** (technical approach, management plan, past performance).
- **Automated scoring engine** for at least three criteria with human override.

2. MVP 2 (Month 4)

- **Advanced AI scoring:** neural-network models trained on historical award data.
- **Blockchain-based immutable audit log** for all reviewer actions and AI decisions.
- **Integrated FAR/OCI rule-engine** for automated compliance checks.

3. MVP 3 (Month 9)

- **Interactive evaluator dashboard:** side-by-side AI vs. human scores, inconsistency alerts.
- **Collaborative review workflows** with threaded comments, digital signature capture.
- **Exportable source-selection documentation package** (sensitivities, findings, scoring rationale).

4. Final Release (Month 18)

- Full end-to-end source-selection SaaS with ATO authorization in AWS GovCloud.
 - Enterprise integrations: e-Business Suite, defense contract databases.
 - Training materials, admin console, sustainment & DevSecOps pipeline.
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4. Technical Approach

1. Data & NLP Pipeline

- Curate labeled corpus of past solicitations and evaluation reports.
- Fine-tune transformer models (e.g., domain-adapted BERT) for factor extraction.
- Implement robust PDF/Word parsers, OCR fallback for scanned docs.

2. AI Scoring & Rule Engine

- Train neural-network scoring models on multi-criteria labels (technical merit, cost realism, performance risk).
- Develop a configurable rule engine encoding FAR sections, solicitation instructions, and OCI constraints to validate AI outputs and human inputs.

3. Blockchain Audit & Security

- Use a private Hyperledger Fabric network to record all evaluation actions immutably.
- Enforce RBAC and multi-factor authentication; encrypt data at rest and in transit per DoD STIGs.

4. User Experience & Workflow

- Apply human-centered design: iterative user feedback sessions with contracting officers and evaluators.
 - Build a web-based dashboard for real-time visibility into scoring distributions, flag resolutions, and evaluation progress.
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5. Work Plan & Milestones

Timeframe	Activities	Deliverable
Months 0–1	Project kickoff; finalize requirements; ingest sample solicitations	Detailed SOW & Data Schema
Day 30	MVP 1 deployment: ingestion + basic NLP + automated scoring	MVP 1 Demonstration Video
Months 2–4	Train advanced scoring models; develop FAR/OCI rule engine; audit blockchain	MVP 2 Beta Release
Months 5–8	Build evaluator dashboard; collaborative workflows; documentation export	MVP 3 Release
Months 9–16	Security hardening; AWS GovCloud ATO prep; final integration & performance	Pre-release Candidate
Months 17–18	Final testing; user training; sustainment planning; Phase II report	Final Deliverables & ATO Pack

6. Key Personnel

Name	Role	Expertise
Dr. Riley Morgan	Principal Investigator	12 yrs AI/ML in defense acquisition; ex-DARPA program lead
Ms. Olivia Chen	Acquisition SME	10 yrs Army source-selection authority; FAR/OCI specialist
Mr. Victor Lee	Lead Data Scientist	8 yrs NLP & neural-network model development
Ms. Sarah Patel	Blockchain Architect	7 yrs Hyperledger, secure ledger design
Mr. Daniel Ortiz	UX/UI & Workflow Designer	9 yrs human-centered design for DoD portals

7. Company Overview

[Company Name] is a small business certified CMMC Level 3 and FedRAMP Ready, specializing in AI-driven defense procurement solutions. We maintain an ITAR-compliant AWS GovCloud environment and a continuous DevSecOps pipeline. Our team has delivered multiple SBIR Phase II and III products to the DoD, achieving rapid ATO approvals and broad user adoption.

8. Facilities & Environment

- **Secure Lab:** Segmented networks, hardware firewalls, SIEM monitoring.
 - **Compute Resources:** GPU clusters for model training; EKS clusters for SaaS deployment.
 - **Toolchain:** Terraform/Ansible for infrastructure; GitLab CI/CD with security scanning.
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9. DP2 Deliverables

1. **Day 30: MVP 1**—basic ingestion, NLP extraction, automated scoring.
 2. **Month 4: MVP 2**—advanced AI scoring, FAR/OCI rule engine, blockchain audit.
 3. **Month 9: MVP 3**—evaluator dashboard, collaboration, exportable documentation.
 4. **Month 18: Production-ready SaaS** with ATO, training, and sustainment plan.
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10. Phase III & Dual-Use Applications

Beyond the Army, SelectAI-SS can be tailored for other federal agencies (e.g., DHS, GSA) and highly regulated sectors (finance, healthcare) to automate RFP evaluation and vendor selection, leveraging its modular AI/rule-engine architecture.